

Total No. of Questions : 8]

SEAT No. :

PD4211

[Total No. of Pages : 3

[6403] 4

T.E. (Civil)

**ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT
(2019 Pattern) (Semester - V) (301004)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of non programmable scientific calculator is allowed.

- Q1) a)** What is Financial Management? Explain “Why construction financial management is different?” **[6]**
- b)** What are the roles and responsibilities of construction finance manager? **[6]**
- c)** Write down the definition of contract and explain: **[5]**

- i) Percentage rate contract
- ii) Item rate contract

OR

- Q2) a)** The following were the expenses on a contract which commenced on 1st January, 2020. **[8]**
- i) Materials purchased = Rs. 4,40,000
 - ii) Direct wages = Rs. 1,28,000
 - iii) Plant issued = Rs. 40,000
 - iv) Overhead = Rs. 17,200
 - v) Material in hand = Rs. 20,000.

The contract price was Rs. 12,00,000 against which the cash received on account of contract was Rs. 6,00,000. The plant has gone 20% depreciation. Prepare the contract account.

- b)** Explain the importance of Financial management in construction. **[4]**
- c)** What is Contract? Explain why it is prepared? **[5]**

P.T.O.

- Q3) a)** Explain in brief the essential elements of budget. **[4]**
b) What is 'Rule of 72'? Write down its formula. **[5]**
 If Mr. 'A' has invested Rs. 20,00,000 at a 8% rate of interest, in how many number of years the invested amount gets doubled?
c) Calculate the Internal Rate of Return (IRR) on the investment in the following cash flow: **[8]**

Year	0	1	2	3	4	5
Cash inflow	1,00,000	+20,000	+30,000	+20,000	+40,000	+40,000

Consider 10% discount rate for first calculation.

OR

- Q4) a)** From the following details relating to two projects A and B, suggest which one is to be accepted using Net Present Value (NPV) method? **[6]**

Particular	Project A	Project B
Estimated Cost (Rs.)	1,10,000	1,10,000
Cash flow in year 1 (Rs.)	31,000	71,000
Cash flow in year 2 (Rs.)	40,000	40,000
Cash flow in year 3 (Rs.)	50,000	40,000
Cash flow in year 4 (Rs.)	70,000	20,000

The company expects a return of 10%

- b)** Write down a detailed note on "Project Expenditure Budget". **[5]**
c) A concrete mixer was purchased at Rs. 9,00,000. Assuming Scrap Value to be Rs. 6,50,000 after 10 years. Calculate the depreciation at the end of 5 years by **[6]**
 i) Straight Line Method
 ii) Constant Percentage Method
- Q5) a)** Write down the significance of working capital in construction company. Enlist any 6 factors considered for determination of working capital. **[6]**
b) Considering 10% contingencies, calculate the working capital from the following data; **[6]**

Particulars	Amount (Rs.)	Particulars	Amount(Rs.)
Bank Balance	2,55,000	Account Payable	2,80,000
Term loan from Bank	2,95,000	Stock of finished goods	1,20,000
Payable unclaimed dividend	55,000	Account receivable	3,00,000
Commission receivable	62,000	Staff PF	80000

- c)** What is classification problem in inventory management? How to solve this problem? Explain any one method. **[6]**

OR

- Q6) a)** Write a note on following: [6]
 i) Types of working capital
 ii) Components and financing resources of working capital
- b)** The yearly requirement of cement by a construction firm is 250 bags. The cost of one bag of cement is Rs. 375 and ordering cost per order is Rs. 100. Assume annual carrying cost for inventory of 20% of average inventory management. Calculate EOQ and number of orders in one year. If the lead time is 5 days calculate order point. [6]
- c)** Explain Safety stock, stating its importance and various cost involved. [6]
- Q7) a)** What is direct tax? Explain any two types of direct tax in India. [6]
- b)** Calculate the property tax for the apartment which used only for residential purpose, from following location data: [6]
 i) No of floors – 14 (Floor factor -1.10)
 ii) Base value – Rs. 370/Sqft
 iii) User category factor – 0.5
 iv) Nature and type of structure – RCC (factor-1)
 v) Age of building – 18 years (Age factor – 0.85)
 vi) Carpet area – 550 Sq ft
 vii) Location – Thane
 viii) Tax rate – 0.775%
- c)** Explain any 6 functions of Reserve Bank of India (RBI). [6]
- OR**
- Q8) a)** What are objectives the of Tax planning? Explain in brief Corporate tax planning. [6]
- b)** Write down the roles and functions of following financial regulatory bodies: [6]
 i) ICRA (Information and Credit Rating Agency)
 ii) SEBI (Security and Exchange Board of India)
- c)** Calculate the payable property tax to be paid on property located in Chennai from following data; [6]
 Plinth area= 400sq. ft.
 Monthly rent notified = Rs. 6 per sq. ft.
 Tax rate = 30%
 Library cess = 15%
 Consider 10% rebate on age of building.